Tutorial 4: K-means and Gaussian Mixtures

From hard assignments to probabilistic clustering

Kangning Cui
City University of Hong Kong (Dongguan)

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How to use this tutorial

- Attempt each problem before reading its solution.
- Write the main calculation or argument clearly.
- Explain what the result means, not only the final number or formula.
- Complete Tasks 1–3 first. The final task is an optional challenge.

Task 1: K-means initialization and k-means++

Consider one-dimensional data:

$$x = \{1, 2, 5, 8, 9\}.$$

We want $K = 2$ clusters.
First use initialization A:

$$\mu_1 = 1, \qquad \mu_2 = 5.$$

1. Perform one assignment step.
2. Compute the initial objective.
3. Update the centroids.

Then consider initialization B:

$$\mu_1 = 2, \qquad \mu_2 = 8.$$

Repeat the same calculation.
Compare the two initializations.

Solution: Initialization A assignment

For initialization A:

$$\mu_1 = 1, \qquad \mu_2 = 5.$$

Distances:

x	1	2	5	8	9
$\lvert x - \mu_1 \rvert$	0	1	4	7	8
$\lvert x - \mu_2 \rvert$	4	3	0	3	4

Therefore:

$$\{1, 2\} \to 1, \qquad \{5, 8, 9\} \to 2.$$

The initial objective is:

$$J_A = (1-1)^2 + (2-1)^2 + (5-5)^2 + (8-5)^2 + (9-5)^2.$$

Therefore:

$$J_A = 26.$$

Solution: Initialization A update

Update centroids:

$$\mu_1 = \frac{1+2}{2} = 1.5.$$

$$\mu_2 = \frac{5+8+9}{3} = \frac{22}{3}.$$

The updated objective becomes:

$$J_A^{new} = (1-1.5)^2 + (2-1.5)^2 + (5-\frac{22}{3})^2 + (8-\frac{22}{3})^2 + (9-\frac{22}{3})^2.$$

Thus:

$$J_A^{new} \approx \frac{55}{6}.$$

The update decreases the objective.

Solution: Initialization B

For initialization B:

$$\mu_1 = 2, \qquad \mu_2 = 8.$$

Distances give:

$$\{1, 2, 5\} \to 1,$$

$$\{8, 9\} \to 2.$$

The initial objective is:

$$J_B = (1-2)^2 + (2-2)^2 + (5-2)^2 + (8-8)^2 + (9-8)^2.$$

Therefore:

$$J_B = 11.$$

Compared with initialization A:

$$J_B < J_A.$$

Initialization B starts from a better solution.

K-means++ intuition

Random initialization may select poor initial centers.
K-means++ chooses new centers with probability proportional to:

$$D(x)^2,$$

where $D(x)$ is the distance to the nearest existing center.
The idea:

far points $\rightarrow$ higher probability of becoming centers.

This encourages:

diverse initial centroids.

K-means++ improves initialization by spreading initial centers across the data.

Task 2: GMM responsibility calculation

A point x may be generated by three Gaussian components.
The component information is:

$$\pi_1 = 0.2, \qquad p(x|z=1) = 0.9,$$

$$\pi_2 = 0.5, \qquad p(x|z=2) = 0.3,$$

$$\pi_3 = 0.3, \qquad p(x|z=3) = 0.4.$$

1. Compute the responsibility of each component.
2. Which component has the largest posterior probability?
3. Explain why the largest π_k does not always produce the largest responsibility.

Solution: GMM unnormalized probabilities

The responsibility is:

$$\gamma_k = \frac{\pi_k p(x|z=k)}{\sum_j \pi_j p(x|z=j)}.$$

First compute the unnormalized scores:

$$s_1 = 0.2 \times 0.9 = 0.18,$$

$$s_2 = 0.5 \times 0.3 = 0.15,$$

$$s_3 = 0.3 \times 0.4 = 0.12.$$

The normalization term is:

$$S = 0.18 + 0.15 + 0.12 = 0.45.$$

Solution: GMM responsibilities

Normalize each score:

$$\gamma_1 = \frac{0.18}{0.45} = 0.40.$$

$$\gamma_2 = \frac{0.15}{0.45} = 0.33.$$

$$\gamma_3 = \frac{0.12}{0.45} = 0.27.$$

Therefore:

$$(\gamma_1, \gamma_2, \gamma_3) = (0.40, 0.33, 0.27).$$

Component 1 has the largest posterior probability.

Solution: interpreting responsibility

The responsibility combines two factors:

prior component probability

and:

how well the component explains the point.

Component 2 has the largest prior:

$$\pi_2 = 0.5,$$

but component 1 has higher likelihood:

$$p(x|z = 1) = 0.9.$$

Therefore component 1 has the largest posterior probability.
This demonstrates:

posterior = prior $\times$ likelihood

after normalization.

Task 3: EM weighted parameter update

After the E-step, one component receives the following responsibilities:

$$\gamma = (0.1, 0.7, 0.8, 0.4)$$

for observations:

$$x = (2, 4, 7, 9).$$

1. Compute the effective membership:

$$N_k = \sum_i \gamma_i.$$

2. Compute the updated mean.
3. Compute the updated mixing weight assuming $n = 4$.
4. Explain why EM uses weighted statistics.

Solution: EM effective membership

The effective membership is:

$$N_k = 0.1 + 0.7 + 0.8 + 0.4.$$

Therefore:

$$N_k = 2.$$

The updated mean is:

$$\mu_k = \frac{0.1(2) + 0.7(4) + 0.8(7) + 0.4(9)}{2}.$$

The numerator is:

$$0.2 + 2.8 + 5.6 + 3.6 = 12.2.$$

Therefore:

$$\mu_k = 6.1.$$

Solution: EM mixing weight and interpretation

The updated mixing weight is:

$$\pi_k = \frac{N_k}{n}.$$

Since:

$$N_k = 2, \qquad n = 4,$$

we obtain:

$$\pi_k = 0.5.$$

The weighted mean differs from the ordinary mean:

$$\frac{2+4+7+9}{4} = 5.5.$$

The reason is that EM does not treat all points equally.
A point with larger responsibility contributes more to the parameter update.

Optional challenge: diagnose a GMM model

A researcher trains two GMM models.
Model A:

$$K = 2, \qquad \text{validation likelihood is stable.}$$

Model B:

$$K = 10, \qquad \text{training likelihood is much higher but validation performance is worse.}$$

1. Which model is more reliable?
2. Explain why training likelihood alone is insufficient.
3. Give two methods for selecting K.

Optional challenge: implement clustering with GMM

Use a synthetic dataset and compare K-means with Gaussian mixture models.

1. Generate a two-dimensional dataset containing several groups.
2. Train a K-means model and a GMM model.
3. Visualize the clustering results.
4. Compare the hard assignments from K-means with the soft responsibilities from GMM.

Tutorial 4 complete

Questions and discussion